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EEL 3135 – Classwork #05

Milestone:

- Complete any 2 questions to receive credit.

Question #1: Given:

$$x(t) = 4e^{j(10t+\pi/6)} + 4e^{-j(10t+\pi/6)}$$

(a) Use Euler's identity to convert $x(t)$ into its real sinusoidal form.

Since inverse amplitude is split in half, we multiply it by two.

$$x(t) = 8\cos\left(10t + \frac{\pi}{6}\right)$$

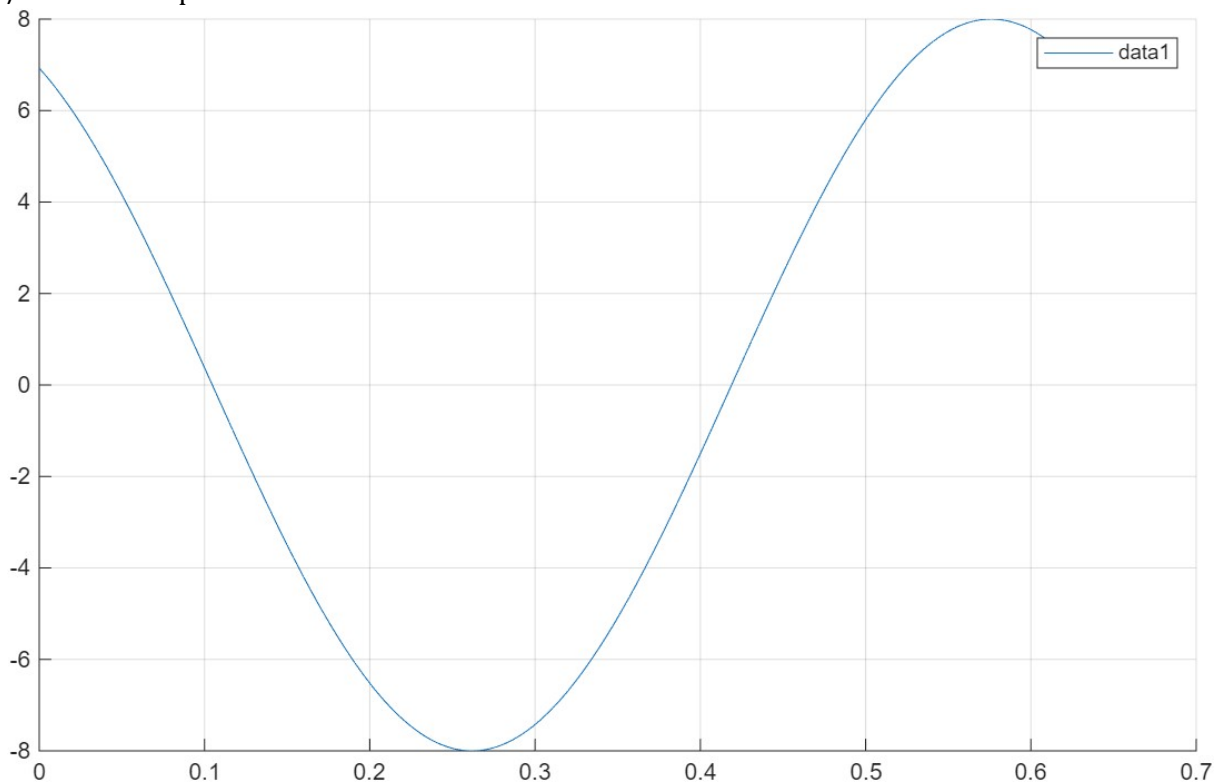
(b) Identify the amplitude, phase shift, and frequency of the resulting cosine wave.

Amplitude: 8

Phase: $\frac{\pi}{6}$

Frequency: $\frac{10}{2\pi}$

(c) Sketch one period of the waveform in Matlab.

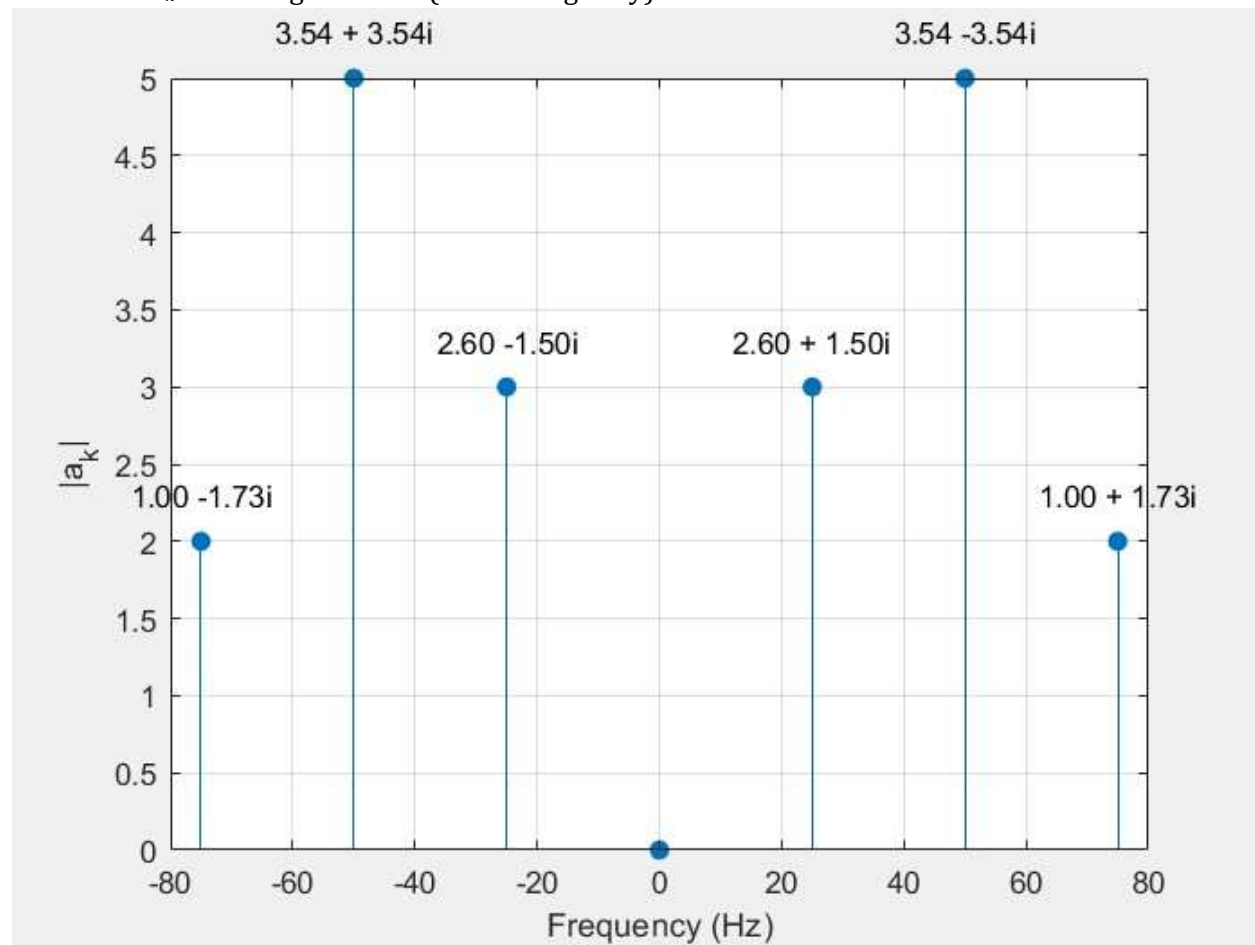


Question #2: You are given the following frequency-domain information for $y(t)$:

$$\omega_0 = 5 \text{ rad/s}, \quad a_0 = 3, \quad a_{\pm 1} = 2e^{\mp j\pi/4}, \quad a_{\pm 2} = 1e^{\mp j\pi/3}$$

- (a) Write $y(t)$ as a sum of complex exponentials.
- (b) Rewrite $y(t)$ using cosine terms (real Fourier series).
- (c) Plot $y(t)$ over one period in MATLAB.

Question #3: The figure below shows the two-sided magnitude spectrum of $x(t)$ with frequency components at $f = \pm 25, \pm 50, \pm 75$ Hz. Each spectral line is labeled with its corresponding complex coefficient a_k in rectangular form (real + imaginary).



(a) Determine the fundamental frequency f_0 and fundamental period T_0 of $x(t)$.

$$f_0 = \text{GCD}(25, 50, 75) = 25 \text{ Hz}$$

(b) Write $x(t)$ in complex exponential form using the coefficients a_k from the spectrum.

For $\pm 75\text{Hz}$:

$$\begin{aligned} & |1.00 - 1.73i|e^{j75*2\pi t} + |1.00 - 1.73i|e^{-j75*2\pi t} \\ &= 1.9982e^{j75*2\pi t} + 1.9982e^{-j75*2\pi t} \end{aligned}$$

For $\pm 50\text{Hz}$:

$$\begin{aligned} & |3.54 + 3.54i|e^{j50*2\pi t} + |3.54 + 3.54i|e^{-j50*2\pi t} \\ &= 5.0063e^{j50*2\pi t} + 5.0063e^{-j50*2\pi t} \end{aligned}$$

For $\pm 25\text{Hz}$:

$$\begin{aligned} & |2.60 - 1.50i|e^{j25*2\pi t} + |2.60 - 1.50i|e^{-j25*2\pi t} \\ &= 3.0017e^{j25*2\pi t} + 3.0017e^{-j25*2\pi t} \end{aligned}$$

$$a_{-3} = 1.9982e^{-j75*2\pi t}$$

$$a_{-2} = 5.0063e^{-j50*2\pi t}$$

$$a_{-1} = 3.0017e^{-j25*2\pi t}$$

$$a_0 = 0$$

$$a_1 = 3.0017e^{j25*2\pi t}$$

$$a_2 = 5.0063e^{j50*2\pi t}$$

$$a_3 = 1.9982e^{j75*2\pi t}$$

(c) Use Euler's formula to rewrite $x(t)$ as a sum of cosines in the time domain.

Total:

$$6.0033\cos(50\pi t - 0.5233) + 10.0126\cos(100\pi t + 0.7854) + 3.9964\cos(150\pi t - 1.0467)$$

Question #4: Given

$$y(t) = 4 \cos\left(12t + \frac{\pi}{3}\right) - 7 \cos\left(24t - \frac{\pi}{6}\right) + 2 \cos(36t),$$

- (a) Evaluate $y\left(\frac{\pi}{12}\right)$.
- (b) Express $y(t)$ in complex exponential form using Euler's formula.
- (c) Sketch the magnitude spectrum of $y(t)$ at the harmonic frequencies $k\omega_0$ using Matlab.